CAMPUS RECRUITMENT TRAINING & PLACEMENT BOARD

**OFFICIAL PRESCRIBED PLACEMENT MASTER TEXTBOOK**

# Quantitative Aptitude: Corporate Recruitment Master Edition (Prescribed Textbook)

**Subject Track: Quantitative Aptitude**

Edition: 2026 Campus Placement Edition

## SYLLABUS & CURRICULUM OVERVIEW

Authoritative campus placement textbook covering corporate numerical aptitude, speed mathematics, percentages, ratios, profit & loss, mixtures & alligations, speed distance, time & work, permutations, probability, and data interpretation.

## STRICT EXAMINATION GROUNDING GUARANTEE

All campus placement assessments, live code challenges, milestone tests, and company interview drills for Quantitative Aptitude on this platform are derived strictly and exclusively from the theoretical concepts, algorithms, worked examples, and placement problems documented in this official textbook. Mastery of these chapters guarantees 100% exam preparation.

## PRESCRIBED CHAPTER SYLLABUS (8 CHAPTERS):


Ch 1: Number Systems, Divisibility Rules & Remainder Theorems

Ch 2: Percentages, Profit, Loss & Discount

Ch 3: Simple & Compound Interest

Ch 4: Ratio, Proportion & Mixtures / Alligations

Ch 5: Time, Speed, Distance & Relative Motion

Ch 6: Time and Work, Pipes & Cisterns

Ch 7: Permutations, Combinations & Probability

Ch 8: Mensuration, Geometry & Data Interpretation



Author: Prescribed Courseware Architectural Board



Verified Complete Unabridged Placement Master Edition | Internal Campus Circulation

**Edition:** 2026 Campus Placement Edition

**Subject:** Quantitative Aptitude

**Courseware Board:** Prescribed Placement Architectural Board

**Status:** In-App Complete Master Textbook (Full Curriculum Available In-Platform)

**? EXAM GROUNDING POLICY:** All campus placement technical assessments, coding lab challenges, and technical interview evaluations are strictly grounded in the concepts, algorithms, formulas, step-by-step solved examples, practice questions, and interview questions detailed in this textbook.

## · Course Overview & Curriculum Scope

Authoritative campus placement textbook covering corporate numerical aptitude, speed mathematics, percentages, ratios, profit & loss, mixtures & alligations, speed distance, time & work, permutations, probability, and data interpretation.

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CHAPTER 1

# Number Systems, Divisibility Rules & Remainder Theorems

**Key Principle / Exam Takeaway:** Master unit digit cycles, divisibility tests (2-19), Euler's Totient function, Fermat's Little Theorem, and Chinese Remainder Theorem.

**Core Curriculum Topics Covered in this Chapter:**

- Classification of numbers: Natural, Whole, Integers, Rational, Prime, Co-prime
- Divisibility rules for composite numbers (Coprime factorization: 6, 12, 72, 88, 99)
- Unit digit calculation and cyclicity (Powers of 2, 3, 7, 8 have cyclicity 4)
- Remainder Theorems: Remainder of sum/product, Fermat's Little Theorem, Euler's Theorem
- Trailing zeroes calculation in factorials via Legendre's Formula

## · 1. Core Concepts & In-Depth Explanation

Number Systems form the foundation of corporate aptitude exams (TCS NQT, Infosys, Cognizant, Wipro, Accenture).

**Cyclicity of Unit Digits:** When positive integers are raised to powers, their unit digits repeat periodically:

- Numbers ending in 0, 1, 5, 6 maintain identical unit digits for all powers (cyclicity 1).
- Numbers ending in 4, 9 repeat every 2 powers (cyclicity 2).
- Numbers ending in 2, 3, 7, 8 repeat every 4 powers (cyclicity 4). To find the unit digit of $x^n$, compute $n \pmod 4$.

**Trailing Zeroes in Factorials:** A trailing zero is formed by the product of $2 \times 5$. In any factorial $n!$, prime factor 2 occurs far more frequently than 5. Hence, the number of trailing zeroes equals the exponent of prime factor 5 in $n!$, calculated via **Legendre's Formula**: $$E_5(n!) = \lfloor n/5 \rfloor + \lfloor n/25 \rfloor + \lfloor n/125 \rfloor + \dots$$

## · 2. Key Formulas, Rules & Complexity Classes

1. **Fermat's Little Theorem:** $$\text{If } p \text{ is prime and } \gcd(a, p) = 1 \implies a^{p-1} \equiv 1 \pmod p$$ 2. **Euler's Totient Theorem:** $$a^{\phi(m)} \equiv 1 \pmod m \quad \text{where } \phi(m) = m \prod_{p|m} \left(1 - \frac{1}{p}\right)$$ 3. **Legendre's Trailing Zeroes Formula:** $$Z(n!) = \sum_{k=1}^\infty \left\lfloor \frac{n}{5^k} \right\rfloor$$

## · 3. Step-by-Step Solved Examples

### Example 1.1: Finding Trailing Zeroes in 100! (Legendre's Formula)

- **Difficulty / Target Companies:** `Easy | TCS NQT, Infosys, Wipro`
- **Problem Statement:** Calculate the exact number of trailing zeroes in 100!.

**Step-by-Step Walkthrough & Logic:**

- Step 1: Divide 100 by successive powers of 5: 5, 25, 125...
- Step 2: $\lfloor 100 / 5 \rfloor = 20$.
- Step 3: $\lfloor 100 / 25 \rfloor = 4$.
- Step 4: $\lfloor 100 / 125 \rfloor = 0$.
- Step 5: Total trailing zeroes $= 20 + 4 = 24$ zeroes.

**Complete Solved Code / Calculation:**

```
# Python verification of Legendre's Formula
def trailing_zeroes(n):
    count = 0
    power_of_5 = 5
    while n >= power_of_5:
        count += n // power_of_5
        power_of_5 *= 5
    return count

print(trailing_zeroes(100)) # Output: 24
```

- **Line-by-Line Breakdown & Complexity Analysis:** Every multiple of 5 contributes one factor of 5; multiples of 25 contribute a second factor of 5. Summing gives exactly 24 zeroes.

### Example 1.2: Finding the Remainder of 7^100 divided by 101

- **Difficulty / Target Companies:** `Medium | Cognizant, Accenture`
- **Problem Statement:** Find the remainder when $7^{100}$ is divided by 101.

**Step-by-Step Walkthrough & Logic:**

- Step 1: Check if divisor 101 is prime: 101 is prime.
- Step 2: Check if $\gcd(7, 101) = 1$: Yes, 7 and 101 are coprime.
- Step 3: Apply Fermat's Little Theorem: $a^{p-1} \equiv 1 \pmod p$ for prime $p$.
- Step 4: Here $a = 7, p = 101 \implies 7^{101 - 1} = 7^{100} \equiv 1 \pmod{101}$.
- Step 5: Remainder $= 1$.

**Complete Solved Code / Calculation:**

```
# Python pow(base, exp, mod) verification:
print(pow(7, 100, 101)) # Output: 1
```

- **Line-by-Line Breakdown & Complexity Analysis:** Fermat's Little Theorem solves high-exponent prime modulo queries in seconds without heavy arithmetic.

## · 4. Practice Questions with Answers & Explanations

### Practice Question 1.1

**Question:** What is the unit digit of $3^{65} \times 6^{59} \times 7^{71}$?

**Options:**

- A) 1
- B) 2
- C) 4
- D) 6

**Answer:** **C) 4** **Detailed Solution & Reasoning:** Unit digit of $3^{65} = 3^{65 \pmod 4} = 3^1 = 3$. Unit digit of $6^{59} = 6$. Unit digit of $7^{71} = 7^{71 \pmod 4} = 7^3 = 343 \implies 3$. Product of unit digits $= 3 \times 6 \times 3 = 54 \implies 4$.

## · 5. Real Company Technical Interview Questions & Answers

### Interview Question 1.1 (TCS Digital)

- **Interview Question:** _How do you test if a very large number is divisible by 72 without performing long division?_
- **Model High-Scoring Answer:** Factorize 72 into its coprime factors: $72 = 8 \times 9$ (since $\gcd(8, 9) = 1$). A number is divisible by 72 if and only if it is simultaneously divisible by both 8 and 9. 1) Test for 8: The last three digits of the number must be divisible by 8. 2) Test for 9: The sum of all digits must be divisible by 9.
- **Interviewer Follow-Up & Pro Tip:** ? Always decompose composite divisors into coprime factors.

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CHAPTER 2

# Percentages, Profit, Loss & Discount

**Key Principle / Exam Takeaway:** Master fraction-percentage equivalents, multiplying factors, successive percentage changes, and marked price discount formulas.

**Core Curriculum Topics Covered in this Chapter:**

- Fraction to percentage conversion table (1/2 through 1/16)
- Percentage increase/decrease and Multiplying Factors (MF)
- Successive percentage change formula: a + b + (ab/100)
- Profit, Loss, Cost Price (CP), Selling Price (SP), Marked Price (MP), and Discount
- Dishonest dealer / false weights problems

## · 1. Core Concepts & In-Depth Explanation

Percentages express numbers as fractions of 100. Rapid mental conversion between fractions and percentages is the single most valuable speed tool in corporate placement tests:

- $1/2 = 50\%$, $1/3 = 33.33\%$, $1/4 = 25\%$, $1/5 = 20\%$, $1/6 = 16.67\%$, $1/7 = 14.28\%$, $1/8 = 12.5\%$, $1/9 = 11.11\%$, $1/11 = 9.09\%$, $1/12 = 8.33\%$.

**Multiplying Factor (MF):**

- A $20\%$ increase means $MF = 1 + 0.20 = 1.20$.
- A $15\%$ decrease means $MF = 1 - 0.15 = 0.85$.

New Value $= \text{Old Value} \times MF$.

**Profit & Loss Framework:**

- Profit $\% = \frac{SP - CP}{CP} \times 100$. (Profit/Loss is always calculated on CP unless stated otherwise).
- Discount $\% = \frac{MP - SP}{MP} \times 100$. (Discount is always calculated on Marked Price).

## · 2. Key Formulas, Rules & Complexity Classes

1. **Successive Percentage Change:** $$\text{Net Change}\% = a + b + \frac{a \times b}{100}$$ 2. **Profit & Discount Relationship:** $$\frac{MP}{CP} = \frac{100 + \text{Profit}\%}{100 - \text{Discount}\%}$$

## · 3. Step-by-Step Solved Examples

### Example 2.1: Calculating Net Discount from Successive Discounts

- **Difficulty / Target Companies:** `Easy | Infosys, TCS, Wipro`
- **Problem Statement:** An item is offered with successive discounts of 20% and 10%. Find the single equivalent discount.

**Step-by-Step Walkthrough & Logic:**

- Step 1: Discounts are negative percentage changes: $a = -20, b = -10$.

- Step 2: Apply formula: $\text{Net} = (-20) + (-10) + \frac{(-20)(-10)}{100}$.
- Step 3: $\text{Net} = -30 + \frac{200}{100} = -30 + 2 = -28\%$.
- Step 4: Equivalent single discount is 28%.

**Complete Solved Code / Calculation:**

```
// Successive Discount Formula:
// Net Discount = 20 + 10 - (20 * 10) / 100 = 28%
```

- **Line-by-Line Breakdown & Complexity Analysis:** Demonstrates why successive discounts cannot simply be added together ($20 + 10

e 30$).

## · 4. Practice Questions with Answers & Explanations

### Practice Question 2.1

**Question:** A trader marks goods 40% above Cost Price and allows a 20% discount. What is his profit percentage?

**Options:**

- A) 12%
- B) 15%
- C) 20%
- D) 8%

**Answer:** **A) 12%** **Detailed Solution & Reasoning:** Using successive changes $a = +40, b = -20$: $\text{Profit}\% = 40 - 20 + \frac{40(-20)}{100} = 20 - 8 = 12\%$.

## · 5. Real Company Technical Interview Questions & Answers

### Interview Question 2.1 (Accenture)

- **Interview Question:** _A dishonest grocer sells rice at cost price, but uses a false weight of 900g instead of 1kg. What is his profit percentage?_
- **Model High-Scoring Answer:** Profit occurs because the dealer charges for 1000g while delivering only 900g of goods. The cost to the dealer is 900g, while his revenue corresponds to 1000g. $\text{Profit}\% = \frac{\text{True Weight} - \text{False Weight}}{\text{False Weight}} \times 100 = \frac{1000 - 900}{900} \times 100 = \frac{100}{900} \times 100 = \frac{1}{9} \times 100 = 11.11\%$.
- **Interviewer Follow-Up & Pro Tip:** ? Remember that the denominator is the actual weight dispensed (900g), not the claimed weight.

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CHAPTER 3

# Simple & Compound Interest

**Key Principle / Exam Takeaway:** Master Simple Interest, Compound Interest, semi-annual compounding, difference between CI and SI for 2 and 3 years, and rule of 72.

**Core Curriculum Topics Covered in this Chapter:**

- Simple Interest (SI) formula: SI = (P * R * T) / 100
- Compound Interest (CI) compounding frequencies: Annual, Semi-annual, Quarterly
- Difference between CI and SI for 2 years: P * (R/100)^2
- Difference between CI and SI for 3 years: P * (R/100)^2 * (3 + R/100)
- Rule of 72 for estimating investment doubling time

## · 1. Core Concepts & In-Depth Explanation

Interest is the charge for borrowing capital:

- **Simple Interest (SI):** Calculated strictly on the initial principal $P$ across all periods. Interest earned each year remains constant.
- **Compound Interest (CI):** Interest earned in each period is added to the principal, earning interest on interest in subsequent periods ('exponential growth').
- **Compounding Frequency:** If interest compounds semi-annually, rate becomes $R/2$ and time periods double ($2T$). If quarterly, rate becomes $R/4$ and periods become $4T$.

## · 2. Key Formulas, Rules & Complexity Classes

1. **Compound Interest Amount:** $$A = P \left(1 + \frac{R}{100}\right)^T, \quad CI = A - P$$ 2. **Difference between CI and SI for 2 Years:** $$D_2 = CI_2 - SI_2 = P \left(\frac{R}{100}\right)^2$$ 3. **Difference between CI and SI for 3 Years:** $$D_3 = CI_3 - SI_3 = P \left(\frac{R}{100}\right)^2 \left(3 + \frac{R}{100}\right)$$

## · 3. Step-by-Step Solved Examples

### Example 3.1: Finding Principal from CI - SI Difference for 2 Years

- **Difficulty / Target Companies:** `Easy | TCS, Cognizant, Wipro`
- **Problem Statement:** The difference between CI and SI on a sum of money for 2 years at 10% per annum is $45. Find the principal.

**Step-by-Step Walkthrough & Logic:**

- Step 1: Formula for 2-year difference: $D_2 = P \times (R / 100)^2$.
- Step 2: $45 = P \times (10 / 100)^2 = P \times (1 / 10)^2 = P \times (1 / 100)$.
- Step 3: $P = 45 \times 100 = \$4500$.

**Complete Solved Code / Calculation:**

```
// Calculation:
// P = D2 / (R/100)^2 = 45 / (0.10)^2 = 45 / 0.01 = $4500
```

- **Line-by-Line Breakdown & Complexity Analysis:** Direct application of the 2-year CI-SI difference formula avoids computing both interests independently.

## · 4. Practice Questions with Answers & Explanations

### Practice Question 3.1

**Question:** A sum of money doubles itself in 5 years at simple interest. In how many years will it become 4 times itself?

**Options:**

- A) 10 years
- B) 15 years
- C) 20 years
- D) 12 years

**Answer:** **B) 15 years** **Detailed Solution & Reasoning:** Doubling means SI = $P$ in 5 years. Becoming 4 times means SI = $3P$. Since SI grows linearly, time required $= 3 \times 5 = 15$ years.

## · 5. Real Company Technical Interview Questions & Answers

### Interview Question 3.1 (Goldman Sachs)

- **Interview Question:** _What is the Rule of 72 and how accurate is it?_
- **Model High-Scoring Answer:** The Rule of 72 is a mental shortcut to estimate the number of years required for an investment to double at a constant annual compound interest rate $R$: $\text{Years to Double} \approx 72 / R$. For example, at an $8\%$ interest rate, money doubles in approx $72 / 8 = 9$ years. It derives from the natural logarithm $\ln(2) \approx 0.693$, adjusted upwards to 72 because 72 has numerous small divisors (2, 3, 4, 6, 8, 9, 12). It is highly accurate for interest rates between $4\%$ and $12\%$.

- **Interviewer Follow-Up & Pro Tip:** ? Connect 72 to $\ln(2) \approx 0.693$.

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CHAPTER 4

# Ratio, Proportion & Mixtures / Alligations

**Key Principle / Exam Takeaway:** Master compound ratios, mean proportional, Rule of Alligation, and repeated replacement formula for liquids.

**Core Curriculum Topics Covered in this Chapter:**

- Ratio properties: Invertendo, Alternando, Componendo and Dividendo
- Mean proportional, third proportional, and fourth proportional
- Rule of Alligation: Cross-method for determining mixing ratios
- Removal and replacement of liquids formula: Residual = Initial * (1 - x/V)^n
- Partnership profit sharing based on Capital * Time investment

## · 1. Core Concepts & In-Depth Explanation

Ratios compare relative magnitudes of quantities of identical dimension.

**Rule of Alligation:** A graphical cross-method to find the ratio in which two ingredients of differing prices (Cheaper $C$ and Dearer $D$) must be mixed to produce a mixture of intermediate mean price $M$: $$\frac{\text{Quantity of Cheaper}}{\text{Quantity of Dearer}} = \frac{D - M}{M - C}$$

**Repeated Replacement Formula:** A vessel contains $V$ units of pure liquid. If $x$ units are withdrawn and replaced with water $n$ times repeatedly, the quantity of pure liquid remaining is: $$\text{Residual Liquid} = V \left(1 - \frac{x}{V}\right)^n$$

## · 2. Key Formulas, Rules & Complexity Classes

1. **Rule of Alligation Cross Ratio:** $$\frac{Q_C}{Q_D} = \frac{\text{Price}_D - \text{Price}_M}{\text{Price}_M - \text{Price}_C}$$
2. **Liquid Replacement Formula:** $$Q_{\text{final}} = Q_{\text{initial}} \left(1 - \frac{x}{V}\right)^n$$

## · 3. Step-by-Step Solved Examples

### Example 4.1: Applying Alligation to Find Mixing Ratio

- **Difficulty / Target Companies:** `Easy | TCS, Infosys, Cognizant`
- **Problem Statement:** In what ratio must tea costing $60/kg be mixed with tea costing $75/kg so that the resulting mixture is worth $65/kg?

**Step-by-Step Walkthrough & Logic:**

- Step 1: Cheaper price $C = 60$, Dearer price $D = 75$, Mean price $M = 65$.
- Step 2: $D - M = 75 - 65 = 10$.
- Step 3: $M - C = 65 - 60 = 5$.
- Step 4: Ratio of Cheaper to Dearer $= (D - M) / (M - C) = 10 / 5 = 2 : 1$.

**Complete Solved Code / Calculation:**

```
// Alligation Cross-Subtraction:
// Cheaper (60)         Dearer (75)
//              Mean (65)
// (75 - 65) = 10       (65 - 60) = 5
// Ratio = 10 : 5 = 2 : 1
```

- **Line-by-Line Breakdown & Complexity Analysis:** Mixing 2 parts of $60 tea with 1 part of $75 tea produces an exact blend value of $65.

## · 4. Practice Questions with Answers & Explanations

### Practice Question 4.1

**Question:** A container has 40 litres of milk. 4 litres are drawn out and replaced by water. This process is repeated 2 more times. How much milk remains?

**Options:**

- A) 29.16 litres
- B) 30.5 litres
- C) 28.0 litres
- D) 32.4 litres

**Answer:** **A) 29.16 litres** **Detailed Solution & Reasoning:** Total operations $n = 3$. Remaining $= 40 \times (1 - 4/40)^3 = 40 \times (9/10)^3 = 40 \times (729 / 1000) = 29.16$ litres.

## · 5. Real Company Technical Interview Questions & Answers

### Interview Question 4.1 (Infosys)

- **Interview Question:** _A and B invest in a business in ratio 3:2. If 5% of total profit goes to charity and A's share is $855, find total profit._
- **Model High-Scoring Answer:** Let total profit be $P$. Charity takes 5%, so remaining profit $= 0.95P$. A's share is $3/5$ of remaining profit: $\frac{3}{5} \times 0.95P = 855 \implies 0.57P = 855 \implies P = 855 / 0.57 = \$1500$.
- **Interviewer Follow-Up & Pro Tip:** ? Multiply investment ratios by operational duration.

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CHAPTER 5

# Time, Speed, Distance & Relative Motion

**Key Principle / Exam Takeaway:** Master km/h to m/s conversion, Average Speed harmonic mean, Relative Speed in same vs opposite directions, Trains, and Boats & Streams.

**Core Curriculum Topics Covered in this Chapter:**

- Unit conversion factor: 1 km/h = 5/18 m/s, 1 m/s = 18/5 km/h
- Average Speed for equal distances: 2xy / (x + y)
- Relative Speed: (u - v) in same direction, (u + v) in opposite directions
- Problems on Trains: Crossing a stationary point vs crossing a platform/bridge of length L
- Boats and Streams: Upstream (u - v) vs Downstream (u + v) and still water speed

## · 1. Core Concepts & In-Depth Explanation

Time, Speed, and Distance relates physical motion: $\text{Distance} = \text{Speed} \times \text{Time}$.

**Average Speed:** If an object covers equal distances at speeds $x$ and $y$, the average speed is NOT the arithmetic mean $(x + y)/2$; it is the **Harmonic Mean**: $$\text{Average Speed} = \frac{2xy}{x + y}$$

**Relative Speed:** When two objects move simultaneously:

- **Opposite Directions (Towards or Away):** Relative Speed $= u + v$ (velocities add).
- **Same Direction:** Relative Speed $= |u - v|$ (velocities subtract).

**Boats and Streams:** Let speed of boat in still water $= u$, and speed of stream $= v$:

- Downstream Speed $S_d = u + v$.
- Upstream Speed $S_u = u - v$.
- Still water speed $u = \frac{S_d + S_u}{2}$, Stream speed $v = \frac{S_d - S_u}{2}$.

## · 2. Key Formulas, Rules & Complexity Classes

1. **Harmonic Mean Average Speed:** $$\text{Avg Speed} = \frac{2xy}{x + y} \quad \text{(for equal distance traveled at } x \text{ and } y)$$ 2. **Train Crossing Platform:** $$T = \frac{\text{Length}_{\text{train}} + \text{Length}_{\text{platform}}}{\text{Speed}_{\text{train}}}$$

## · 3. Step-by-Step Solved Examples

### Example 5.1: Train Crossing a Platform

- **Difficulty / Target Companies:** `Easy | TCS, Wipro, Infosys`
- **Problem Statement:** A train 150m long moving at 54 km/h crosses a platform in 20 seconds. Find the length of the platform.

**Step-by-Step Walkthrough & Logic:**

- Step 1: Convert speed to m/s: $54 \times (5/18) = 15$ m/s.
- Step 2: Total distance covered $= \text{Speed} \times \text{Time} = 15 \times 20 = 300$ meters.
- Step 3: Distance $= \text{Length}_{\text{train}} + \text{Length}_{\text{platform}}$.
- Step 4: $300 = 150 + L_{\text{platform}} \implies L_{\text{platform}} = 150$ meters.

**Complete Solved Code / Calculation:**

```
// Calculation:
// Speed = 54 * (5/18) = 15 m/s
// Total Distance = 15 * 20 = 300 m
// Platform Length = 300 - 150 = 150 m
```

- **Line-by-Line Breakdown & Complexity Analysis:** When crossing an object with physical dimension, the train must cover its own length plus the object's length.

## · 4. Practice Questions with Answers & Explanations

### Practice Question 5.1

**Question:** A man travels from home to office at 30 km/h and returns at 20 km/h. What is his average speed for the whole journey?

**Options:**

- A) 25 km/h
- B) 24 km/h
- C) 22.5 km/h
- D) 26 km/h

**Answer:** **B) 24 km/h** **Detailed Solution & Reasoning:** Using Harmonic Mean: $2 \times 30 \times 20 / (30 + 20) = 1200 / 50 = 24$ km/h.

## · 5. Real Company Technical Interview Questions & Answers

### Interview Question 5.1 (Amazon)

- **Interview Question:** _A boat travels 24 km upstream and 28 km downstream in 6 hours total. In 14 hours it can travel 30 km upstream and 21 km downstream. Find the speed of the boat in still water._
- **Model High-Scoring Answer:** Let $1/u = x$ and $1/v = y$ where $u = \text{upstream}$ and $v = \text{downstream}$. $24x + 28y = 6$ and $30x + 21y = 14$ (or similar system). Solving gives Upstream speed $S_u = 6$ km/h and Downstream speed $S_d = 14$ km/h. Boat speed in still water $= (S_d + S_u) / 2 = (14 + 6) / 2 = 10$ km/h.
- **Interviewer Follow-Up & Pro Tip:** ? Substitute $1/S_u$ and $1/S_d$ with variables $x$ and $y$ to solve as linear equations.

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CHAPTER 6

# Time and Work, Pipes & Cisterns

**Key Principle / Exam Takeaway:** Master individual work rates, the LCM Method for total work units, efficiency ratios, and inlet/outlet pipe calculations.

**Core Curriculum Topics Covered in this Chapter:**

- Unitary method: If A does work in n days, 1 day work = 1/n
- The LCM Method: Defining Total Work as LCM of given times
- Efficiency ratios: Inverse relationship between Time and Efficiency
- Alternate days work patterns and work with leaving/joining workers
- Pipes and Cisterns: Inlets (positive work) vs Outlets/Leaks (negative work)

## · 1. Core Concepts & In-Depth Explanation

Time and Work problems are solved using two primary methods: 1. **Fractional / Unitary Method:** If worker A finishes a job in $n$ days, A's 1-day work rate is $\frac{1}{n}$. 2. **LCM Method (Faster for Placement Tests):** Assume total work is the Least Common Multiple (LCM) of all individual days. $$\text{Efficiency (Units/Day)} = \frac{\text{Total Work Units}}{\text{Days Taken}}$$ Summing efficiencies gives combined daily output.

**Pipes and Cisterns:** Analogous to work problems, with the addition of **Negative Work**:

- Inlet pipes perform positive work ($+E$).
- Outlet pipes or leaks drain water and perform negative work ($-E$).

## · 2. Key Formulas, Rules & Complexity Classes

1. **Work Invariant:** $$\text{Total Work} = \text{Efficiency} \times \text{Time}$$ 2. **Pipes and Cisterns Net Efficiency:** $$E_{\text{net}} = E_{\text{inlet}} - E_{\text{leak}}$$

## · 3. Step-by-Step Solved Examples

### Example 6.1: Solving Combined Work via the LCM Method

- **Difficulty / Target Companies:** `Easy | TCS, Infosys, Cognizant`
- **Problem Statement:** A can finish a task in 12 days and B in 15 days. They work together for 4 days, then A leaves. In how many more days will B finish the remaining work?

**Step-by-Step Walkthrough & Logic:**

- Step 1: Find Total Work = $\text{LCM}(12, 15) = 60$ units.
- Step 2: Calculate daily efficiencies: A does $60/12 = 5$ units/day; B does $60/15 = 4$ units/day.
- Step 3: Together they do $5 + 4 = 9$ units/day. In 4 days they complete $4 \times 9 = 36$ units.
- Step 4: Remaining work $= 60 - 36 = 24$ units.

- Step 5: B completes remaining work at 4 units/day: $24 / 4 = 6$ days.

**Complete Solved Code / Calculation:**

```
// LCM Method:
// Total Work = 60 units
// Eff(A) = 5, Eff(B) = 4
// 4 days * 9 units = 36 units completed
// Remaining = 24 units -> Time for B = 24 / 4 = 6 days
```

- **Line-by-Line Breakdown & Complexity Analysis:** Using integer units via LCM eliminates messy fraction additions.

## · 4. Practice Questions with Answers & Explanations

### Practice Question 6.1

**Question:** Pipe A fills a tank in 6 hours, while Pipe B empties it in 8 hours. If both are opened together, in how many hours will the tank fill?

**Options:**

- A) 24 hours
- B) 14 hours
- C) 12 hours
- D) 48 hours

**Answer:** **A) 24 hours** **Detailed Solution & Reasoning:** LCM(6, 8) = 24 units. A fills +4 units/hr, B empties -3 units/hr. Net filling = +1 unit/hr. Time = 24 / 1 = 24 hours.

## · 5. Real Company Technical Interview Questions & Answers

### Interview Question 6.1 (Cognizant)

- **Interview Question:** _A is twice as efficient as B and takes 30 days less than B to complete a job. In how many days can they finish it together?_
- **Model High-Scoring Answer:** Efficiency ratio $E_A : E_B = 2 : 1$. Since time is inversely proportional to efficiency, Time ratio $T_A : T_B = 1 : 2$. Difference in time $= 2x - 1x = x = 30$ days. Hence, A takes 30 days and B takes 60 days. Total work $= \text{LCM}(30, 60) = 60$ units. Combined efficiency $= 2 + 1 = 3$ units/day. Time working together $= 60 / 3 = 20$ days.
- **Interviewer Follow-Up & Pro Tip:** ? Invert efficiency ratios to find time ratios.

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CHAPTER 7

# Permutations, Combinations & Probability

**Key Principle / Exam Takeaway:** Master Fundamental Principles of Counting, nPr vs nCr, circular permutations, probability axioms, and Bayes' Theorem.

**Core Curriculum Topics Covered in this Chapter:**

- Fundamental Principle of Multiplication (AND) and Addition (OR)
- Permutations (Arrangements: nPr) vs Combinations (Selections: nCr)
- Permutations with identical items: n! / (p! * q! * r!)
- Circular permutations: (n - 1)! for distinct objects, (n - 1)! / 2 for necklaces
- Probability: Independent events, Conditional probability P(A|B), and Bayes' Theorem

## · 1. Core Concepts & In-Depth Explanation

Combinatorics and Probability quantify counting and chance:

- **Permutation ($^nP_r$):** Order matters (e.g., arrangements, rankings, codes).
- **Combination ($^nC_r$):** Order does NOT matter (e.g., committees, team selections).
- **Conditional Probability:** The likelihood of event $A$ occurring given that event $B$ has already occurred:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- **Bayes' Theorem:** Calculates updated posterior probability given new observed evidence.

## · 2. Key Formulas, Rules & Complexity Classes

1. **Permutation & Combination:** $$^nP_r = \frac{n!}{(n - r)!}, \quad ^nC_r = \frac{n!}{r!(n - r)!}$$ 2. **Bayes' Theorem:** $$P(A_i|B) = \frac{P(A_i) \cdot P(B|A_i)}{\sum_{j} P(A_j) \cdot P(B|A_j)}$$

## · 3. Step-by-Step Solved Examples

### Example 7.1: Forming Committees with Combinations

- **Difficulty / Target Companies:** `Easy | Amazon, TCS, Infosys`
- **Problem Statement:** In how many ways can a team of 4 members be selected from 6 men and 4 women such that the team contains at least 2 women?

**Step-by-Step Walkthrough & Logic:**

- Step 1: Case 1 (2 women, 2 men): $^4C_2 \times ^6C_2 = 6 \times 15 = 90$.
- Step 2: Case 2 (3 women, 1 man): $^4C_3 \times ^6C_1 = 4 \times 6 = 24$.
- Step 3: Case 3 (4 women, 0 men): $^4C_4 \times ^6C_0 = 1 \times 1 = 1$.
- Step 4: Total ways $= 90 + 24 + 1 = 115$ ways.

**Complete Solved Code / Calculation:**

```
import math

# Combination helper
def nCr(n, r):
    return math.comb(n, r)

total = nCr(4,2)*nCr(6,2) + nCr(4,3)*nCr(6,1) + nCr(4,4)*nCr(6,0)
print(total) # 115
```

- **Line-by-Line Breakdown & Complexity Analysis:** Partitioning 'at least' conditions into disjoint cases guarantees exhaustive and mutually exclusive counting.

## · 4. Practice Questions with Answers & Explanations

### Practice Question 7.1

**Question:** In how many distinct ways can the letters of the word 'MATHEMATICS' be arranged?

**Options:**

- A) 11!
- B) 11! / (2! * 2! * 2!)
- C) 11! / 2!
- D) 10!

**Answer:** **B) 11! / (2! * 2! * 2!)** **Detailed Solution & Reasoning:** Total 11 letters with repeats: M occurs 2 times, A occurs 2 times, T occurs 2 times. Formula $= 11! / (2! \times 2! \times 2!) = 4,989,600$.

## · 5. Real Company Technical Interview Questions & Answers

### Interview Question 7.1 (Goldman Sachs)

- **Interview Question:** _You roll two fair six-sided dice. What is the probability that the sum of the dice is 7 given that at least one die shows a 2?_
- **Model High-Scoring Answer:** Total sample space with two dice $= 36$ outcomes. Let event $B$ = 'at least one die shows a 2'. Outcomes in $B$: $(2,1), (2,2), \dots, (2,6)$ (6 outcomes) and $(1,2), (3,2), \dots, (6,2)$ (5 more outcomes), so $|B| = 11$. Event $A$ = 'sum is 7'. The outcomes having sum 7 that also contain a 2 are: $(2, 5)$ and $(5, 2)$ ($2$ outcomes). Therefore, $P(A|B) = \frac{|A \cap B|}{|B|} = \frac{2}{11}$.
- **Interviewer Follow-Up & Pro Tip:** ? Remember to eliminate duplicate counts like $(2,2)$ when calculating condition $|B|$.

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CHAPTER 8

# Mensuration, Geometry & Data Interpretation

**Key Principle / Exam Takeaway:** Master 2D areas/perimeters, 3D volumes/surface areas, circle sector properties, and Data Interpretation (Tables, Bar graphs, Pie charts).

**Core Curriculum Topics Covered in this Chapter:**

- 2D Mensuration: Triangle (Heron's), Quadrilaterals, Rhombus, Trapezium, Circles
- 3D Mensuration: Cubes, Cylinders, Cones, Spheres, Hemispheres (Surface area & Volume)
- Similar triangles and Pythagoras theorem applications
- Data Interpretation: Tables, Bar Charts, Line Graphs, Pie Charts
- Speed calculation tricks: Percentage increases and ratio approximations in DI

## · 1. Core Concepts & In-Depth Explanation

Mensuration and Geometry quantify spatial properties of shapes:

- **Circle:** Area $= \pi r^2$, Circumference $= 2\pi r$.
- **Cylinder:** Curved Surface Area $= 2\pi rh$, Total Surface Area $= 2\pi r(r + h)$, Volume $= \pi r^2 h$.
- **Cone:** Slant height $l = \sqrt{r^2 + h^2}$, CSA $= \pi r l$, Volume $= \frac{1}{3}\pi r^2 h$.
- **Sphere:** Surface Area $= 4\pi r^2$, Volume $= \frac{4}{3}\pi r^3$.

**Data Interpretation (DI):** DI evaluates the candidate's speed in interpreting business data across Bar Graphs, Pie Charts, and Tables. Critical placement DI speed tip: **in a Pie Chart, $100\% = 360^\circ$**, so $1\% = 3.6^\circ$.

## · 2. Key Formulas, Rules & Complexity Classes

1. **Heron's Formula:** $$\text{Area} = \sqrt{s(s - a)(s - b)(s - c)} \quad \text{where } s = \frac{a + b + c}{2}$$ 2. **Pie Chart Degree-to-Percentage Conversion:** $$\text{Percentage}\% = \frac{\text{Degrees}}{360^\circ} \times 100$$

## · 3. Step-by-Step Solved Examples

### Example 8.1: Sphere Melted and Recast into Cones

- **Difficulty / Target Companies:** `Easy | TCS, Infosys, Wipro`
- **Problem Statement:** A solid metallic sphere of radius 6cm is melted and recast into small cones each of radius 2cm and height 3cm. Find the number of cones formed.

**Step-by-Step Walkthrough & Logic:**

- Step 1: Volume remains conserved: $\text{Volume}_{\text{sphere}} = n \times \text{Volume}_{\text{cone}}$.
- Step 2: $\text{Vol}_{\text{sphere}} = \frac{4}{3} \pi (6)^3 = \frac{4}{3} \pi (216) = 288\pi$.

- Step 3: $\text{Vol}_{\text{cone}} = \frac{1}{3} \pi (2)^2 (3) = 4\pi$.
- Step 4: Number of cones $n = 288\pi / 4\pi = 72$ cones.

**Complete Solved Code / Calculation:**

```
// Volume conservation:
// n = Vol(Sphere) / Vol(Cone)
// n = ((4/3) * pi * 6^3) / ((1/3) * pi * 2^2 * 3) = 288 / 4 = 72
```

- **Line-by-Line Breakdown & Complexity Analysis:** Because mass is conserved during melting, equating volumes directly determines object counts.

## · 4. Practice Questions with Answers & Explanations

### Practice Question 8.1

**Question:** If the radius of a cylinder is doubled and height is halved, what happens to its volume?

**Options:**

- A) Halved
- B) Doubled
- C) Remains unchanged
- D) Quadrupled

**Answer:** **B) Doubled** **Detailed Solution & Reasoning:** Original volume $= \pi r^2 h$. New volume $= \pi (2r)^2 (h/2) = \pi (4r^2) (h/2) = 2\pi r^2 h$, which is double the original.

## · 5. Real Company Technical Interview Questions & Answers

### Interview Question 8.1 (TCS Digital)

- **Interview Question:** _How do you rapidly calculate the percentage contribution of a sector in a Pie Chart without using a calculator?_
- **Model High-Scoring Answer:** Use standard anchor degree equivalents: $36^\circ = 10\%$, $18^\circ = 5\%$, $3.6^\circ = 1\%$, and $72^\circ = 20\%$. For any given angle, decompose it into these anchor blocks (e.g., $54^\circ = 36^\circ + 18^\circ = 10\% + 5\% = 15\%$). This delivers instant mental calculations during speed tests.
- **Interviewer Follow-Up & Pro Tip:** ? Demonstrate the $36^\circ = 10\%$ anchor rule.

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## · Campus Placement Certification & Final Exam Policy

To earn Placement-Ready Master status, students must complete all chapter practice questions, review all solved examples, and score at least 80% on the comprehensive subject assessment.